



Detection of Fake Reviews Using Group Model

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Abstract

Reviews of product or stores exist extensively in online e-commerce platform which is important for customers to make decisions. For economic reasons some dishonest people are employed to write fake reviews which is also called “opinion spamming” to promote or demote target products and services. Previous researches have made use of text similarity, linguistics, rating patterns, graph relationship and other behaviors for spammer detection. They mainly utilized product review list while it is difficult to find fake reviews by glancing over product reviews in time-descending order. Meanwhile there exists lots of useful information in the list of reviews of reviewers and relationships between reviewers when reviewers commonly reviewed the same stores. We propose the concept of review group and to the best of our knowledge, it’s the first time the review group concept is proposed and used. Review grouping algorithm is designed to effectively split reviews of reviewer into groups which participate in building novel grouping models to identify both positive and negative deceptive reviews. Several new features which are language independent based on group model are constructed. Additionally, we explore the collusion relationship between reviewers to build reviewer group collusion model. Evaluations show that the review group method and reviewer group collusion models can effectively improve the precision by 4%–7% compared to the baselines in fake reviews classification task especially when reviews are posted by professional review spammers.

Keywords Fake review detection · Opinion spamming · Review group detection · Reviewer group · Reviewer collusion

1 Introduction

People have been active for years in online platform like [amazon.com](https://www.amazon.com), [eBay.com](https://www.eBay.com), [taobao.com](https://www.taobao.com), [yelp.com](https://www.yelp.com), [dianping.com](https://www.dianping.com) etc. and many of them are willing to share opinions about the product they buy or service they get. Meanwhile, some dishonest people write fake reviews just for economic reasons or get credit from the website. Some other people who we called “pseudo professional reviewer” may gather together via offline interaction or instant messaging software to complete a task assigned by store employers. These review spamming behaviors have badly influence customer purchase decisions [1] and the fame of the platform like [yelp.com](https://www.yelp.com), [dianping.com](https://www.dianping.com), [amazon.com](https://www.amazon.com) etc.

It’s a difficult thing to identify a review to be fake or not just by browsing the review list of products because fraudulent reviews are always mixed together when presenting by the website. Meanwhile, it’s easier to identify fake reviews by browsing the review list of a reviewer in time-ascending order. Reviews of reviewers can be grouped based on the assumption that reviews in a group are likely to have similar credibility which express the possibility to be fake or not. Spammers also try to cooperate to write fake reviews on the targeting stores (or products), the cooperating relations and collusion behaviors between reviewers can be found when they write reviews on the same stores.

We propose algorithms to segment reviews of reviewers to groups and build models based on group. These group models are used to classify reviews to be fake or not or used to predict the credibility of each review in regression process. Mukherjee [2] has used the concept “group” to spot reviewer group, we extend the concept of group to include review group and reviewer group. Due to the fact that some “pseudo professional reviewers” cooperate with each other to spam the target store or product, we proposed reviewer relation computation model to compute the collusion index of each reviewer. The main contribution of the paper includes:

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- Proposed a review group detection algorithm to split reviews of reviewer into groups which helps to create different types of features hidden in reviewer review list.
- Proposed method to find reviewer collusion group and modeled the reviewer relationship, built reviewer collusion index models which discovered the reviewer collusion behaviors in review network.
- Built models based on review group and reviewer group to classify reviews as fake or not, or to predict the credibility of a review using regression. These models covered both the review and reviewer group behaviors.
- Experiment and evaluations show that the proposed algorithm and models are effective both in fake review detection and review credibility prediction.

2 Related work

Researches have been done since Jindal [3–5] first proposed the concept “opinion spam”. On the whole, the review spam (opinion spam) detecting method can be classified to three types: supervised learning, unsupervised methods and semi-supervised learning [6]. Supervised learning method [3, 5, 7] is used when labelled data are provided. It has relatively good performance given the right feature and labelled training data. The advantage of the supervised learning is that it can utilize the latent character of labelled data and make use of priori knowledge to conduct the classification task. The disadvantage of the method is it requires rather quantity of labelled data which may require more effort of labor in labelling especially when human labelling is needed. Unsupervised methods use unlabeled data to find unseen relationships between instances independent of a class attribute. Semi-supervised learning is a combination of the two and uses a few labeled instances in combination with a large number of unlabeled instances to train a classifier [8].

Researches in Jindal [3] using supervised learning build classifiers utilizing duplicate or near-duplicate reviews as fake reviews and the rest as non-fake reviews. Ott [7] and Law [9] utilize review content, part-of-speech (POS), LIWC text features and language model to find deceptive opinion spam, but did not consider the user behavior which is very useful to detect fake review and reviewer. Unsupervised method has been used to detect group spammers [2, 10] and review burstiness [11] and behavioral footprints [4]. Mukherjee [2] Proposed a method to find candidate reviewer groups and build relation models based on relationships among groups. Our work still uses the concept “group” but extend the concept of group to including both the relation between reviewers and the review group of reviewers. Fei et.al. [11, 12] exploit the burstiness nature of reviews and co-bursting to identify review spammers.

In a wide field, recent studies in opinion spamming are done in social networks [13–16]. Opinion spamming detection of social networks is quite different from that in product and store review spamming. Lack of user relation links like follow-friend relation makes detection of review spamming a challenging work. Other related research includes trust of website [17] and user trust in social networks [18–20].

Before we propose the review group detection method and reviewer group model, we first introduce the previous work on review and reviewer spamming detection to be as benchmark.

Content similarity-based features are widely used in [2, 3, 11]. Researchers have been using the n-gram [7, 21–23] and POS features to model the content of reviews frequently to study the spamming of reviews. Similar contents of reviews have bigger probability to be spam. Here we incorporate three different levels of content similarity which include: (a) self-content similarity in review (b) content similarity between pairwise reviews of reviewer and (c) content similarity between reviewer and other reviewers. The three levels of content features cover inner similarity of a review, inner similarity of a reviewer, and similarity between reviewers about content. Additionally, we add review length as another content-based feature.

Behavior features are important indicators of reviewers’ spamming activity. Researches have been done based on behaviors which include rating behavior [2, 4], user profile characteristics etc. Sentiment analysis has also been incorporated in the detection process [24–27]. Review sentiment polarity is computed to be utilized as import feature to identify fake reviews. Comparison of existed method and our proposed method are listed in Table 1.

Graph based review spammer group detection techniques are studied in recent years. Wang [29] first proposed heterogeneous review graph to capture the relationship among reviewers, reviews and stores that the reviewers have reviewed. Wang Z. [30] exploited the topological structure of the underlying reviewer graph which reveals that co-review collusiveness and modeled spammer groups as bi-connected graphs. Graph based method can sufficiently utilize the relationship between reviewers and mine the relying information, but the disadvantage is that graph can be very big and building big network or graph can be very time costing. Deep learning-based method has been used in many domains but for the reason that deep learning requires lots of labelled data, hence the method has not been widely researched in fake review detection domain.

3 Review group detection

Browsing the review list of a reviewer is more helpful to fake review detection than browsing the review list of a product or store. The review list of reviewer contains more information about review habits, spamming abnormal behavior and

regularities. Previous works in spamming detection mainly focus in fake reviewer detection and considers all the reviews of fake reviewers to be fraudulent which is too arbitrary and coarse-grained. In fact, even a fake reviewer or “pseudo professional reviewer” sometimes posts real reviews and at some other time post fake reviews. Because the fake reviewer may post real reviews to cover up his spamming behavior or post reviews to product he really bought and experienced for personal needs. Hence, spamming behaviors and normal behaviors often occurred intensively. Adjacent reviews often have similar credibility when they are similar in behavior and content. We group similar reviews together from the first review of the reviewer. The distance formula between a review and group can be defined as follows:

$$\text{distance}(g, re) = \sqrt{\sum_{i=1}^{|v|} \omega_i (v_{gi} - v_i^e)^2} \quad (1)$$

ω is the weight of review posting time, review rating, store category and review content emotion etc. Initially the weight of time and rating are given a relatively high value (0.35) because of its importance and category and emotion weight is given a relatively low value (0.15). v_{gi} is the vector of group i and v_i^e is the vector representation of review re .

The computation of distance implies that the larger the distance between a review and group is, the lower possibility of the review being clustered into this group. The distance measure considers several factors which include time distance, rating distance, category distance and emotional distance. Time distance refers to the posting date similarity between a review and group. Rating distance denotes the rating distance between a review and group. Category distance check if the category of the store associated with the review is the same with that of the group. Emotion factor takes the emotion balance of each review into consideration. Emotion can be positive, negative or

neutral by means of computing the emotional tendency. Then each review has an index of positive, negative and neutral emotion.

We designed an algorithm (Algorithm 1) to detect the groups of reviews of each reviewer. All reviews are parted into many groups with different size. The group detection algorithm is listed below:

Algorithm 1 Sequential Group Detection

Input: R: The reviewer set

RE: set of reviews of each reviewer.

α : max threshold of date distance

δ : distance threshold

Output: group $G_{i,j}$

Initialization: $m=1$;

$G_{1,j} = \{1^{\text{st}} \text{ review of reviewer } j\}$;

currentGroup $cg = G_{1,1}$;

```

1: for each reviewer j in R
2:   for each review REi in RE
3:     if cg is null
4:       cg = {REi}
5:     if (date_between(REi, cg) > α)
6:       output cg
7:       create new group cg = G|Gj|+1,j = {}
8:       continue
9:     Compute distance β utilizing Eq. (1)
10:    If (β < δ)
11:      cg = cg ∪ {REi};
12:      Update centroid of cg;
13:      continue
14:    else
15:      output cg
16:      cg = G|Gj|+1,j = {REi}
17:      continue
18:    end for
19:  end for
  
```

Table 1 Comparison of existed methods and our proposed method

Authors	Key Concept	Features	Learners	Result
Jindal and Liu [3]	Text duplication	review, reviewer and product related	Logistic regression	AUC:78%
Ott et al. [7]	Text Categorization	LIWC (Linguistic Inquiry and Word Count), Bigram	SVM, Naïve Bayes	Precision:89%
Mukherjee et al. [4]	Behavioral footprints	Behavioral feature	SVM, clustering	Precision:74.6%
Mukherjee et al. [2]	Spammer group	Spammer group feature	SVM rank	User agreement result
Shojaee et al. [28]	Stylometric	Lexical and Syntactical	SVM	F1:84%
Rout et al. [25]	Content similarity and sentiment polarity	Sentiment score, linguistic features and unigram	SVM Naïve Bayes Decision Tree	Accuracy:88% Accuracy:91% Accuracy:92%
Proposed approach	Review group detection and collusion	Review group and reviewer collusion related	SVM Naïve Bayes Random Forest	Precision:92% Precision:87% Precision:96%

$G_{i,j}$ denotes the i^{th} group which belongs to reviewer j . RE_i denotes review i in review set RE of a reviewer. In algorithm 1, line 3 and 4 initialize the current group with the current review when current group is empty. Lines 5 to 8 consider the review date difference between the current group and current review, if the difference is big enough (than the parameter α), then the review would be put in another new group. In lines 9 to 17, first the distance between review and current group is computed, then if the distance surpasses the threshold β , the review is added to the current group, meanwhile, centroid of current group would be updated; Otherwise, new group is constructed.

The review group algorithm will be used in section 5 for review group features construction.

4 Reviewer group collusion model

For profit reasons, some people may be employed to cooperate with each other to collude in posting fake reviews to target store or product. These behaviors of cooperating to post reviews on targeting stores or products can influence the rating and ranking of stores they reviewed in a short time period. And if a group of reviewers work together only once to promote or to demote a product of store, it's hard to detect them based on their collective behavior [2] and review content. To intuitively understand the existence of collusion behavior from groups of reviewers in real environments of review web site, we crawled review data of stores from different domain which include hotel, personal care & services and movie in dianping.com which is a professional review web site. The dataset statistics is listed in Table 2.

The data mining technique frequent itemset mining (FIM) [31] is used to detect groups of reviewers who have ever reviewed at least four or more (called support number) stores or products with other reviewers. FPGrowth [32] algorithm is used which is generally the fastest and most memory efficient algorithm compared with the early Apriori, AprioriTID [31] algorithm. We define the event of n reviewers posts reviews on the same k stores (or product) as $\text{Transaction}_{i,k}$. For example, five reviewers post reviews on the same four stores as $\text{Transaction}_{5,4}$. Through the scanning of the dataset of

reviews, we can get a number of such transactions or groups. Intuitively, the bigger of n and k is, the more probabilities of $\text{Transaction}_{i,k}$ considered to be a candidate spammer group because the chance of many reviewers review the same lots of stores is relatively small.

Figures 1, 2, 3 present two kinds of histogram statistics from the dataset in Table 2 in three different domains of professional review web site dianping.com. Figures 1a, 2a and 3a are histogram distributions of transactions with different sizes of reviewer groups in each domain. They represent how many groups (transactions) are found when five, six, seven, or even more than ten reviewers reviewing the same several stores. Figures 1b, 2b, 3b are histogram distributions of transactions with different size of store groups in the three domains. They demonstrate how many candidate groups (transactions) there are when they post reviews on the same four to more than ten stores in each domain.

It is clear from Fig. 1a of distribution in hotel domain that the amount of transactions of seven to ten reviewers who post reviews on the same stores (four stores at least) is rather large. 15,294 transactions(groups) at five of x-axis value are found which means there are different 15,294 reviewer groups of different combination when five reviewers reviewed the same four or more hotels. Note that the detected candidate reviewer groups or transactions may be intersected which cause the amount of groups rather high when i and k is big in $\text{Transaction}_{i,k}$. It's interesting in Fig. 1a that the number of transactions increases as the number of reviewers in transactions gets bigger, but when the number of reviewers is bigger than ten, the number of groups gets smaller. This may be caused by the fact that there are more groups of reviewers participating in spamming task when the group size varies from six to ten. Though there are less groups with the number of reviewers changing from eleven to eighteen, the probability of these groups to be spamming is becoming high. For example, there are 2609 groups of reviewers with each group of fourteen reviewers and 138 groups of reviewers with each group of sixteen reviewers targeting at least the same four hotels.

The Fig. 1b depicts the distribution of transactions (groups) when reviewers reviewed different number of hotels. It is obvious that most groups (at least five reviewers per group) reviewed only four or less hotels. There are 1830 groups of

Table 2 Dataset of reviews from dianping.com

Data category	#reviews	#reviewers	Average #review per reviewer	Average #review per store	#store
Hotel	89,741	74,589	1.2	386	232
Personal Care & Services	67,953	50,708	1.34	240	211
Movie	75,453	67,093	1.12	2794	27

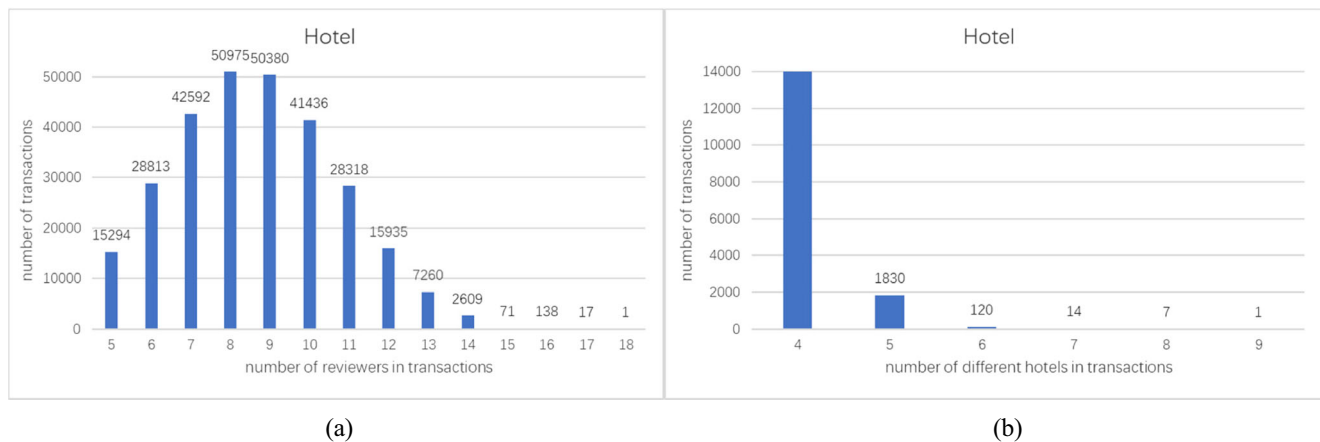


Fig. 1 **a** Transaction distribution over different number of reviewers when they reviewed at least four hotels. **(b)** Histogram distribution of transactions with different size of store groups in hotel domain

reviewers who reviewed the same five hotels and 120 groups targeting the same six hotels which are more likely to be spammer groups. The dataset of movie domain is comprised of reviews from reviewers targeting different movie theaters. From the Fig. 2b we can see that the number of groups of people who reviewed the same five (or less) cinemas are really large. While there are much less groups of people who reviewed the same six or more cinemas.

From all these figures in the three domains, it can be concluded that the bigger the group is and the more stores they reviewed on, the more probabilities they cooperate to be group spammers. Meanwhile, groups of reviewers reviewing the same stores does not mean absolutely the group is a spammer group and the reviewers are spammers. For example, in movie domain, there are indeed many groups of reviewers once visited the same cinemas, but they may not spammers for the reason that they rated the cinemas at different time with different rating.

From analysis of the three datasets, these detected groups are possible spammer groups. But considering all these reviewers in the group to be spammer directly can

wrongly classify innocent reviewers who review the store in the group by accident. Reviewers have different probabilities to be spammer even they are in the same group, cause though they reviewed the same stores, they may differ in rating behaviors and posting dates and other behaviors or review content.

We designed a method to evaluate the relationship between reviewers in the same group and compute the level of collusion of reviewers when they post reviews on the same stores. The method is implemented follow these steps:

- (1) Using FIM (Frequent Item Mining) technique to find frequent item set (candidate group) in which reviewers reviewed the same several stores set just as what we do in the previous part.
- (2) For each candidate group, compute the relationship between every two reviewers in the group. If the two reviewers cooperate to spam on stores and have similar behaviors, the relation value between them will be high. The relationship can be computed as follows:

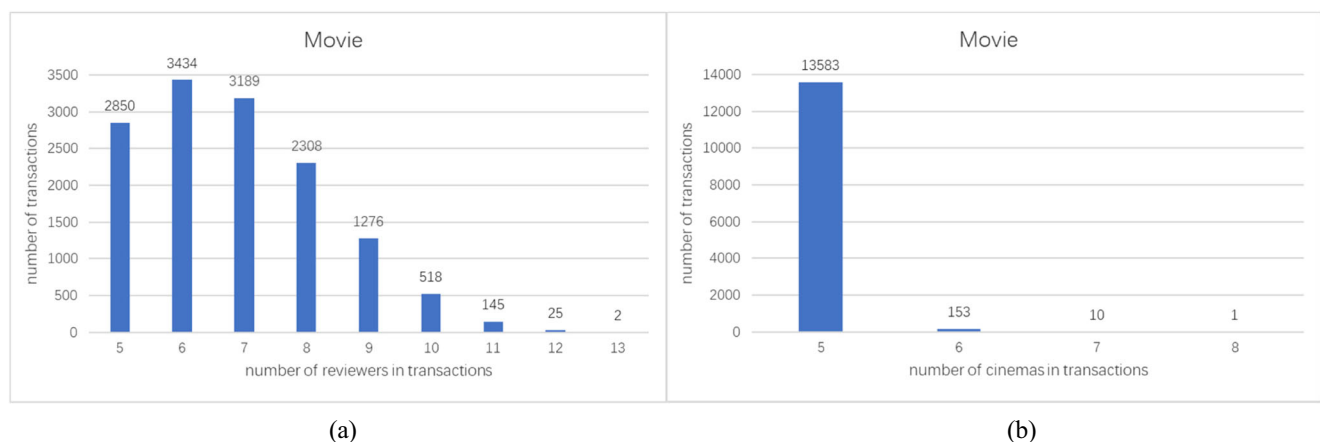


Fig. 2 **a** Histogram distribution of transactions with different size of reviewer groups in movie domain. **(b)** Histogram distribution of transactions with different size of store groups in movie domain

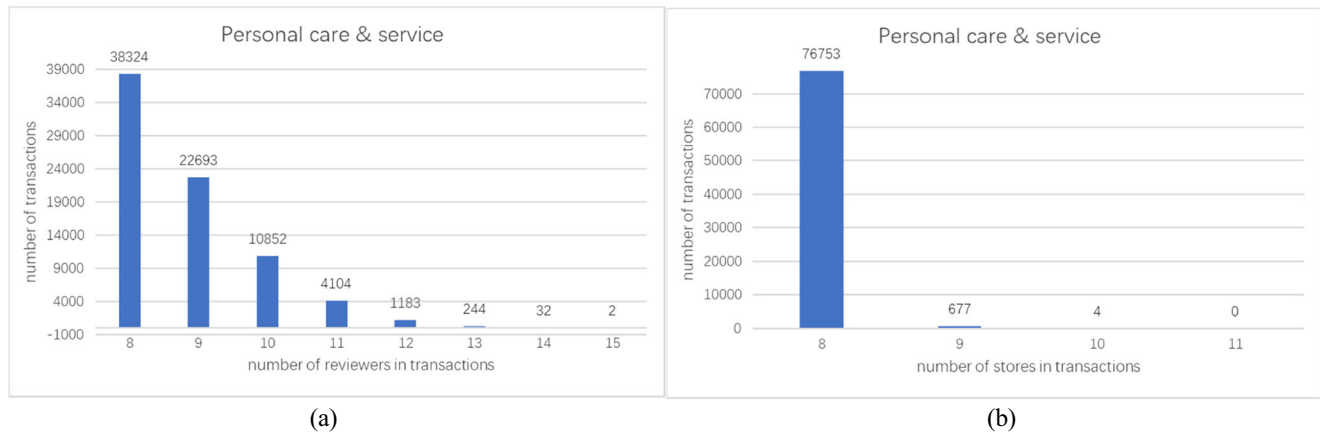


Fig. 3 **a** Histogram distribution of transactions with different size of reviewer groups in personal care & service domain. **(b)** Histogram distribution of transactions with different size of store groups in personal care & service domain

- a) Compute the rating difference between the two reviewers on reviews targeting the same stores. The formula is defined as follows:

$$\text{ratingRelation}(u_1, u_2) = 1 - \frac{1}{|S|} \sum_{i=1}^{|S|} \alpha * \text{ratingDiff}(u_1, u_2)_i \quad (2)$$

where S is the store set which reviewer u_1 and u_2 commonly reviewed. $|S|$ is the size of S . ratingDiff is function of rating difference between reviews from the two reviewers targeting the store i . α is the coefficient to make the $\text{ratingDiff} \in [0, 1]$. In the experiment dataset, the rating varied from one star to five stars, then α is set to one fourth. If reviewers post multiple reviews to the store i , average rating of the reviews of reviewer is used for rating difference computation. If the rating relation equals 1, it implies that the rating behavior of the two reviewers is similar, rated five stars both on stores they reviewed for instance.

- b) Compute the posting date relationship between the two reviewers on reviews targeting the same stores. The formula is defined as follows:

$$\text{dateRelation}(u_1, u_2) = 1 - \frac{1}{|S|} \sum_{i=1}^{|S|} \text{dateDiff}(u_1, u_2)_i \quad (3)$$

$$\text{dateDiff}(u_1, u_2)_i = \frac{2}{\left(1 + e^{\frac{-\sqrt{x}}{2}}\right)} - 1 \quad (4)$$

where x is the date difference between two reviews from the

corresponding reviewers targeting the store i , ranging from zero to several days. When the posting date between two reviews is the same, the dateDiff value equals zero and when they differs too much (like more than two month), the dateDiff will be approaching one. If reviewers post multiple reviews to the store i , minimum date difference between reviews from the two reviewers is used for date difference computation.

- c) Compute relationship between reviewers in group by multiply the results using ratingRelation and dateRelation function in step a) and b)

$$\text{relation}(u_1, u_2) = \text{ratingRelation}(u_1, u_2) * \text{dateRelation}(u_1, u_2) \quad (5)$$

- d) Compute the relation value of reviewer u by averaging all the relation between u and its neighbors (formula 5) in the reviewer group.

$$\text{relation}(u) = \frac{1}{|G|-1} \sum_{i=1}^{|G|-1} \text{relation}(u, u_i) \quad (6)$$

where $|G|$ is the candidate group size.

Relation value of reviewer u records the possibility of u cooperating with other reviewers intentionally. If the relation value is high, reviewer u has relatively similar rating and posting date behavior with other reviewers in the same group. In addition to relation value of reviewer, the group size and number of stores reviewers reviewed commonly should also be considered for spammer collusion index computation. We combined the relation and

group related factor together to compute the index of reviewer r who have collusion behaviors with other reviewers targeting the same stores. The index of reviewer r is computed as follows (formula 7):

$\text{Collusion}(r)$ is the index of reviewer r who have collusion behavior with other reviewers targeting the same stores.

$$\text{collusion}(r) = \text{relation}(r) * \frac{\sum_t |\text{itemset}_{r,t}| \times |\text{support}_{r,t}|}{\sum_r \sum_t |\text{itemset}_{r,t}| \times |\text{support}_{r,t}|} \quad (7)$$

$\text{itemset}_{r,t}$ is set of reviewer ids who have reviewed the same three or more stores. $\text{support}_{r,t}$ is the support number of the itemset which means the reviewers in itemset have a number of $|\text{support}|$ common reviewed stores.

We create reviewer collusion related features using different support numbers which include 4,6,8,10,12 to express different degrees of collusion behavior. When the support number becomes higher, the reviewers cooperate to review more stores which means the probability of collusion is high too.

5 Review and reviewer group model construction

5.1 Reviewer group size related review group model

Dishonest Reviewers often post many reviews that are similar in posting time, rating preferences, category distribution, content emotional distribution etc. to increase their credibility or to promote the stores that employ them or for other illegal reasons. Bigger group usually means more fake reviews posted with the same intension and target which would save their time and enhance efficiency. We demonstrate three review group size related features which include the average reviewer group size (ARGZ), max reviewer group size (MRGZ), and big group size ratio (BGSr).

to cover the group size domain. We normalize ARGZ and BGSr to $[0,1]$.

$$\text{ARGZ}(r) = \frac{|\text{RE}_r|}{|G_r|} \quad (8)$$

$|G_r|$ is the number of groups of reviewer r , $|\text{RE}_r|$ is the number of reviews of reviewer r .

$$\text{BGSr}(r) = \frac{|\text{BG}_r^\lambda|}{|G_r|} \quad (9)$$

$|\text{BG}_r^\lambda|$ is the number of big group of reviewer r based on the parameter λ which defines the number of reviews per group as a big group. In the experiment, we set λ to five which is medium in size.

5.2 Inner group behaviors of review group model

5.2.1 Inner group content similarity

The content of reviews in the same group is similar when spammers copy their reviews among themselves. So, when a review is largely similar to the content of the group, it is more possible to be faked or spammed. The content similarity between review re and group g is defined as:

$$\text{IGCS}(re, g) = \max_{re \in g} (\cos(re, re_g)) \quad (10)$$

where g is the group which contains reviews re . re_g is the review set in g except the review re itself, $\cos$ is the cosine similarity function.

5.2.2 Inner group time density

If a group contains several reviews with the same posting date which means a “high time density”, the group is more likely to be spammed. We model inner group time density as follows:

$$\text{IGTD}(re, g) = \frac{|t_{re,g}|}{|g|} \quad (11)$$

$$\text{IGTD}(g) = \frac{|t_g|}{|g|} \quad (12)$$

where $|t_{re,g}|$ is the number of reviews in group g that has the same posting time with review re . $|t_g|$ is the number of different posting time in group g . $|g|$ denotes the size of group g .

5.2.3 Inner group category density

Professional spammers usually post reviews with the same category when they are given an order to promote the same series of stores like hotels. Reviews on same category with high frequency are abnormal and the reviewers are suspected to be employees of the company. Lower inner group category density reflects review preferences with diversity that normal reviewers have. Inner group category density between review re and group g is defined as:

$$\text{IGCD}(re, g) = \frac{|\text{RE}_{re,g}^c|}{|g|} \quad (13)$$

where $|\text{RE}_{re,g}^c|$ is number of reviews which has the same category c with review re in group g . $|g|$ denotes the size of group g .

5.2.4 Inner group store similarity

Similar to inner group category density, multiple reviews targeting the same or very similar store is suspicious when they are in the same group because this situation usually does not happen by accident but on purpose. Sometimes spammers post reviews to the store of the subbranch of the same brand. We model the similarity of stores in group as follows:

$$IGSS(re, g) = \max_{re \in g, i} (sim(s_{re}, s_{i, g})) \quad (14)$$

where s_{re} is the store which review re is targeted on, and $s_{i, g}$ is store i of group g . We omit the situation of similarity of two same stores which is in vain.

5.2.5 Inner group rate diversity

The rate diversity of groups of reviews can also indicate spamming. At an extreme case, the spammer rates all 5 stars to the store he/she reviewed with no rate diversity. The standard deviation is a measure that is used to quantify the amount of variation or dispersion of a set of data values, thus variance of review ratings of group can be modeled as:

$$IGRD(g) = \sqrt{\frac{1}{n} \sum_{j=1}^n (pr_g - \overline{pr_g})^2} \quad (15)$$

pr_g denotes the proportion of every star rating. Here, we use the five-star rating mechanism and n is set to five. $\overline{pr_g}$ is the average of every pr_g . The inner group rate diversity value attains 0 when all the ratings of reviews are the same like all 5-star rating or all 4-star rating.

Based on the review group model, nine features are constructed for the fake reviews detection task as is shown in Table 3.

5.3 Reviewer group collusion feature construction

As is mentioned in section 4, a series of reviewer collusion features are constructed based on the reviewer group collusion model using different support number of 4, 6, 8, 10, 12 which means that reviewers review at least the same four or more products (stores). We compute the reviewer collusion value using formula (7). As is shown in Table 4, five features based on user collusion behavior are constructed to cover different level of reviewer collusion behaviors. Some other support level of features can be designed like features of support of 3, 5, 7, 9 etc. We select the even support number for simplicity which are described in Table 4.

6 Experiments and evaluation

6.1 Datasets and evaluation metrics

One of the major challenges in this area of research is availability of gold standard datasets [34]. Very few datasets with spamming labeled are open to researchers. Ott. [7] published a dataset with gold standard deceptive opinions gathered using Amazon Mechanical Turk (AMT) in hotel domain. But the dataset is too small which contains only 1600 opinions total and employed AMTs are not genuine reviewers.

when posting reviews. In addition, paid crowdsourcing like AMT can only imitate the content of genuine reviews but they do not reflect actual behavioral and psychological state of mind of fake reviewers [34] which is rather important in fake review detection. Other datasets of product ratings seldom have the spamming labels and are lack of reviewer behavior information. So, we crawled a dataset of reviews and reviewers from dianping.com which is a popular opinion sharing website as it provides us a real scenario for data analysis. First, we search the stores of express inn in a given

Table 3 Feature construction based on review group model

Feature number	Group related feature	Description
F1	Group contents similarity	Max contents similarity in group
F2	Group Time Density	Number of different Date in group / group size
F3	Group Category Density	Number of Category in group / group size
F4	Group Store Similarity	Max Store Similarity in group
F5	Group Rating Diversity	Mean rating variance in group
F6	Group Size	Group size / max group size
F7	Average Group size	Number of reviews in group / number of groups of current reviewer
F8	Max Group Size	Max number of reviews in group
F9	Big Group Ratio	Number of big groups / number of groups of current reviewer

Table 4 Construction of reviewer collusion feature of different support level

Feature number	Collusion related feature	Description
F10	Collusion Support 4	Collusion model with support number of 4
F11	Collusion Support 6	Collusion model with support number of 6
F12	Collusion Support 8	Collusion model with support number of 8
F13	Collusion Support 10	Collusion model with support number of 10
F14	Collusion Support 12	Collusion model with support number of 12

city which is prone to be spammed for economic reasons, and collect all the reviews of reviewers who have reviewed hotels. Due to the large number of reviewers and corresponding reviews in our data, it would have taken too much time for human judges to assess all the reviewers in a short time. We thus only randomly selected 20 reviewers which contain 4189 reviews to label. Two researchers who are familiar with online shopping and rating are involved in the labelling process. They are given the same dataset of 20 reviewers with all of their reviews and metadata of reviewers such as date of register, number of fans, number of flowers got from other reviewers etc. Table 5 shows the statistical information of labeled dataset. Annotators label the credibility of each review with a float number varying from 0 to 1 instead of binary labeling of spam and non-spam since people are often uncertain about the truth of review. Annotators rate 0 or 0.1, 0.2 when he/she is confident that the review is spam and rate 1 or 0.9, 0.8 when he/she is confident that the review is real after browsing the review text content, ratings, posting date and other historical review information. Finally, average rating of the two annotators are computed as final annotation of reviews. The credibility of reviews can be used in classification or regression process. We split the dataset into two part: spam and non-spam with threshold of 0.5 of credibility in classification.

For the classification evaluation metrics, we use the precision (P in Table 7), recall (R in Table 7), F1 measure (F1 in Table 7) for classification evaluation and use correlation coefficient and MAE for regression process. Precision, Recall and F1 measure are defined as follows:

$P = TP / (TP + FP)$, $R = TP / (TP + FN)$, $F1 = 2P * R / (P + R)$ which TP means number of instances that are true positive, FP means number of instances that are false positive, FN means number of instances that are false negative.

6.2 Evaluation using classification

Since we have labeled the training data with credibility value between 0 and 1, we consider each review as spam when the credibility is less than 0.5 and take each review as non-spam when the credibility is labeled as 0.5 and above. We select five popular machine learning classifiers in weka [35] which include Ada Boost, Bagging, Logistic Regression, Naïve Bayes and Random Forest and perform 10-fold cross validation. Table 6 describes reviewer behavior related features used in previous research [24, 33].

In the classification experiment as is shown in Table 7, five different classifiers are used to evaluate the performance of the four feature sets described in Table 3, Table 4, Table 6 and including N-gram and part of speech (POS) features. N-gram and part of speech (POS) features are commonly used in fake review detection task [7, 9, 33, 34]. Since the reviews in dataset of dianping.com are written in Chinese and there isn't any space between words, we first run the word segmentation tool IKAnalyser and perform the POS tag analysis. Results in Table 6 show that N-gram and POS feature set perform worst in spam detection task which is similar to the results in [33]. This illuminates that content-based methods like unigram or bigram and POS feature perform poor in fake review detection task. Spammers will imitate real reviewers and write reviews similar to them with different ratings (high rate for promotion purpose and low rate for defame).

The BM feature set in Table 7 includes features F15-F26 which represent reviewer basic history meta information and behavior related feature like review density related feature in our previous work [24]. Results of BM feature set shows that behavior and reviewer basic meta information especially the behavior feature can boost the efficiency up to more than 20%

Table 5 Labeled dataset of reviews

dataset	#reviews	#reviewers	Average #review per reviewer	#store	#review (spam)	#review (non-spam)	#store category	#group per reviewer	#review per group
dianping.com (Hotel)	4189	20	209	3658	1974	2215	18	8.6	24.3

Table 6 Behavior feature commonly used in previous research

Feature number	Behavior related feature	Description
F15	Good ratio	Good rating number / number of reviews
F16	Bad ratio	Number of bad rating / number of reviews
F17	User review number ratio	Number of user review / max user review number
F18	Review length [33]	Words count per review
F19	Rating variance [24]	Mean variance delta of user reviews
F20	Review number per day	Average review number / #active date
F21	Favorite review ratio	Number of favorite / number of reviews of current reviewer
F22	Focus review ratio	Number of focus / number of reviews of current reviewer
F23	Fans Review ratio	Number of fans / number of reviews of current reviewer
F24	Category density [24]	Number of reviews in current category of review / number of reviews of reviewer
F25	Review density [24]	Number of reviews in current store / number of reviews of reviewer
F26	Time density [24]	Number of different review date / number of reviews of reviewer

than N-Gram and POS feature set. This illuminated that behavior related features contain more information than content of the reviews in fake review detection task. The review group behavior and reviewer group collusion feature set (BC) contains the feature F1-F14 described in Table 3 and Table 4 which are constructed by means of group model in section 5. We take the “NPBS” feature set as the standard comparing to the BC feature sets. Results in Table 7 show improvement of BC than NPBS up to about 7% in precision, recall and F1 measure when SVM classifier is used. BC features achieve an increase of performance at about 4.5%–4.7% in F1 measure by utilizing the Bagging and random forest classifier and get 3.8% improvement by the Naive Bayes classifier. Classifier of Bagging and Random Forest do the best compared to other classifiers in almost all the feature sets. Obviously, group related features captured nature of the fake review and reviewer.

Figure 4. shows the performance of four main feature sets test when using five classifiers. The review group and reviewer collusion model can be effectively applied in supervised learning and get a rather good performance, in fact the group and collusion model can also be used in an unsupervised learning way such as review clustering.

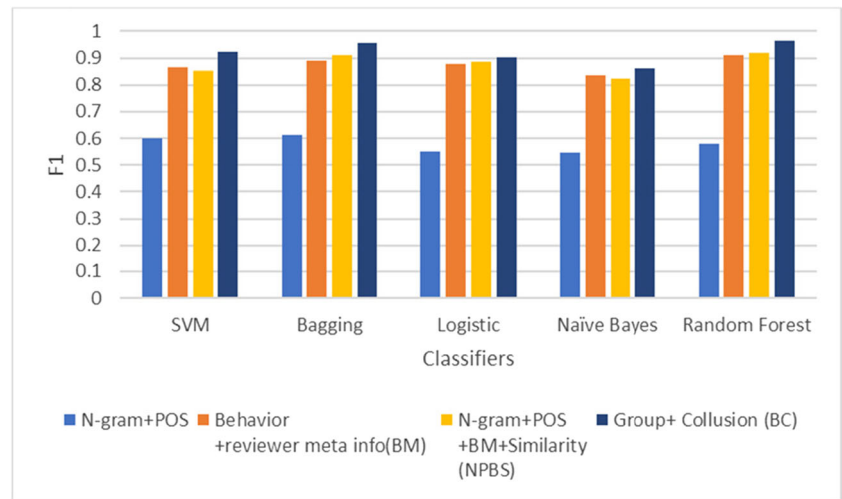
6.3 Regression evaluation

Due to the reason that people sometimes are uncertain on deciding whether the review is fake or not, we label the review with float number ranging from 0 to 1 to represent the credibility of each review. Thus, the newly proposed review group model and reviewer collusion behavior model can be evaluated by regression method. In Figs. 5 and 6, the “all feature” set contains all the feature

Table 7 Comparison of classification between different feature composition using classifier. P, R, F1 denote precision, recall and F1 measure respectively

Classifier Feature	SVM			Bagging			Logistic			Naïve Bayes			Random Forest		
	P	R	F1	P	R	F1	P	R	F1	P	R	F1	P	R	F1
N-gram+POS [33, 34]	0.6	0.6	0.6	0.611	0.611	0.611	0.550	0.553	0.551	0.548	0.548	0.548	0.577	0.579	0.578
Behavior +reviewer meta info (BM) [33]	0.868	0.86	0.864	0.89	0.89	0.89	0.877	0.876	0.876	0.837	0.836	0.836	0.909	0.909	0.909
N-gram+POS +BM + Similarity (NPBS)	0.857	0.850	0.853	0.911	0.910	0.910	0.888	0.886	0.887	0.826	0.825	0.825	0.917	0.917	0.917
Group + Collusion (BC)	0.924	0.924	0.924	0.955	0.955	0.955	0.897	0.907	0.902	0.876	0.851	0.863	0.964	0.964	0.964
Improvement of (BC) than (NPBS) (%)	6.7	7.4	7.1	4.4	4.5	4.5	1.0	2.1	1.5	5.0	2.6	3.8	4.7	4.7	4.7

Fig. 4 F1 of different classifier with four types of feature sets



of N-gram, POS, behavior and reviewer meta information and recommended group features. The “no group” feature set contains the features in “all feature” except the group related feature. “only group” feature set contains only the group related feature of F1 to F14. From the results of regression in Fig. 5, we can see that by using the group model, the correlation coefficient of “all feature” with group model reach high to more than 0.95 which is higher than the situation when no group feature is imported. The “only group” feature set achieves better results with less features than the “no group” feature set when two regression method (bagging and random forest) are used while the same result occurred when SVM is applied. The “only group” feature set performs not very well utilizing the linear regression because of its simplicity and less features. The mean absolute error (MAE in Fig. 6) of regression process is rather low when bagging and random forest with group feature (all

feature) are used. Linear regression performs worst for its simplicity.

6.4 Group parameter evaluation

Some parameters are to be configured to build review group, one of which is the date threshold α of building groups. The value of parameter α defines the number of days a group has to be closed and need to open a new group. We test different value of date threshold α of building groups on five classifiers. It is shown in the Figs. 7–8 that bagging and random forest perform the best in precision (likewise in recall and F1 measure) whatever parameter value of threshold α is. But the precision, recall and F measure is decreasing when threshold α changes from 0 to 30 or more for bagging, random forest and SVM. This means when the threshold is 0, it performs the best in evaluation. The reason may be that our method can effectively identify useful review groups without manually

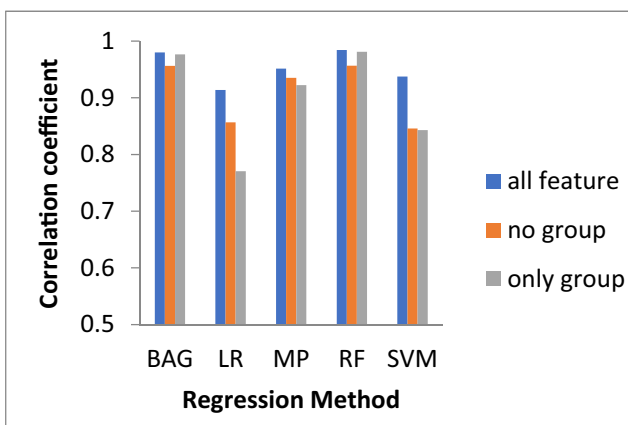


Fig. 5 Correlation coefficient of five different regression method with different feature sets. BAG, LR, MP, RF, SVM is abbreviations of regression method bagging, logistic regression, multilayer Perceptron, random forest and SVM

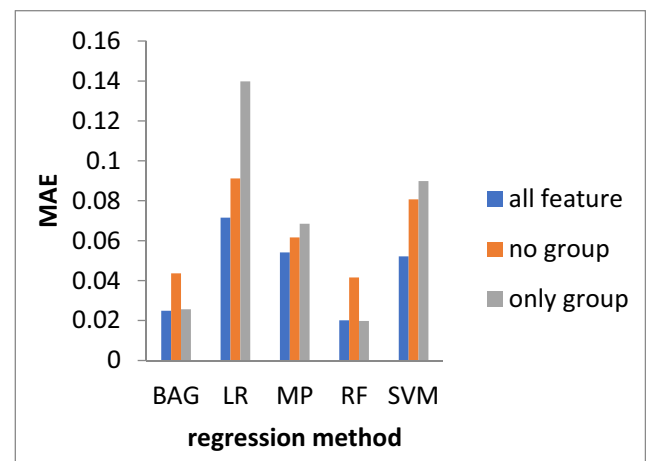


Fig. 6 MAE of five different regression method with different feature Sets

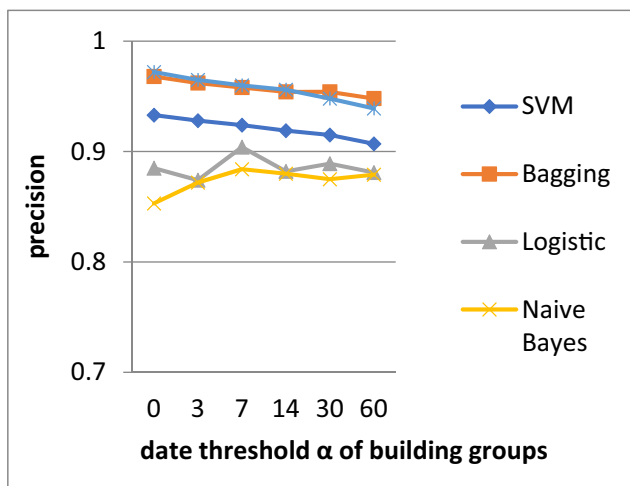


Fig. 7 Precision of five different regression method with different feature sets for group parameter evaluation

specify any threshold of dividing groups. The performance of Logistic and Naïve Bayes method rises when date threshold α of building groups changes from zero to seven and then drop when the threshold α becomes bigger than 14. It can be concluded that reviews of reviewer tend to have the same credibility when they are date related. We can also see that performance changed not much with different date threshold using the same classifier which imply that the performance does not only depend on date threshold but the group distance computation model.

Figure 8 demonstrates the MAE (mean absolute error) of each classifier given different date threshold α in building groups. Still bagging and random forest perform the best and got the lowest MAE. For logistic and naïve Bayes, the MAE level is lower down to bottom when the threshold is 7 and goes up again when the threshold goes up. Generally, date

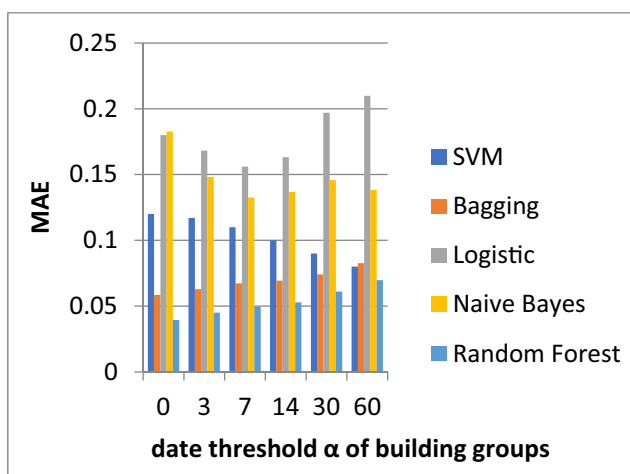


Fig. 8 MAE of five different regression method with different feature sets for group parameter evaluation

threshold α in building groups gets a rather good performance at the date 7. But when the parameter α is bigger than 14, the performance drops down which means that reviews that posted two weeks ago are not likely to be in the same group which is consistent with our intuition. So, in the main experiment shown in Table 6., we select 7 as the alpha threshold to get the best result.

7 Conclusion

Review spamming is becoming a serious problem to the development of e-commerce and constantly increasing fake reviews make normal users difficult to find the right product or store to deal with. According the group characteristic of reviews of reviewer, we propose novel review grouping method and models to identify latent fake reviews. The review grouping algorithm can effectively split reviews of reviewer into groups which participate in building new model of review spamming detection. Additionally, we explore the reviewer group collusion behavior between reviewers to build group collusion model. Experiments and evaluations show that the review group method and reviewer group collusion method and relevant models are easy to implement and can effectively identify fake reviews especially those posted by professional review spammers. Future works include application of review group model using an unsupervised method and do more work in reviewer collusion behavior via community detection technology.

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